4 (a)	acceleration proportional to displacement (from a fixed point) <u>or</u> a = - $\omega^2 x$ with a, ω and x explained	**M1**	
	and directed towards a fixed point <u>or</u> negative sign explained	**A1**	**[2]**
(b)	for s.h.m., $a = (-)\omega^2 x$	**B1**	
	identifies ω^2 as $A\rho g/M$ and therefore s.h.m. (may be implied)	**B1**	
	$2\pi f = \omega$	**B1**	
	hence $f = \frac{1}{2\pi}\sqrt{\frac{Apg}{M}}$	**A0**	**[3]**
(c) (i)	$T = 0.60$ s <u>or</u> $f = 1.7$ Hz	**C1**	
	$0.60 = (2\pi\sqrt{M})/\sqrt{(\pi \times \{1.2 \times 10^{-2}\}^2 \times 950 \times 9.81)}$	**C1**	
	$M = 0.0384$ kg	**A1**	**[3]**
(ii)	<u>decreasing</u> peak height/amplitude	**B1**	**[1]**